Lesson 2 worksheet: Make It Yours: Light Patterns

Name: _____ Date: _____ **Grown-up who checked my circuit:** _____

> **Same circuit as Lesson 1. Don't take it apart.** Keep the USB cable unplugged until a grown-up has checked it. Today only the code changes. Your teacher has put the new `main.py` on the ESP32. You share one laptop with Thonny on it.

1. Predict (look at the code in Thonny with your teacher first)

The top of the program says `RED_S = 5`, `GREEN_S = 5` and `YELLOW_S = 2`. A **variable** is a name that stores a value. In `set_lights(r, y, g)` the numbers are red, yellow, green: 1 = on, 0 = off.

Question	My guess
Will the lights do anything different from Lesson 1?	
Which light does `set_lights(0, 1, 0)` turn on?	
And `set_lights(0, 0, 0)`?	
With `RED_S = 8`, how long is one full cycle?	

2. Check the circuit (USB unplugged, tick each box)

Light	ESP32 pin → hole	Resistor	Long leg (+)	Short leg (−)	Black wire	✔
Red	D25 → b7	d7 to g7	i7	i8	j8 → − rail	☐
Yellow	D26 → b15	d15 to g15	i15	i16	j16 → − rail	☐
Green	D27 → b23	d23 to g23	i23	i24	j24 → − rail	☐

☐ One long black wire joins the **blue − rail** to the GND pin next to D13.

3. Observe (plug in, then use the shared laptop)

Count the seconds: red ____ s, green ____ s, yellow ____ s. Same as Lesson 1? ☐ yes ☐ no

In Thonny, press **Run**. Copy the first three messages from the **Shell**:

1. _____ 2. _____ 3. _____

4. Make it yours: design a pattern on paper, then type it in with your teacher

Phase	Message for `print("...")`	`set_lights(_, _, _)`	Seconds
1		(___ , ___ , ___)	
2		(___ , ___ , ___)	
3		(___ , ___ , ___)	

5. Explain (use your own words)

1. Why write `time.sleep(RED_S)` and not `time.sleep(5)`? _____
2. Why must `def set_lights` come **before** `while True:`? _____
3. The ESP32 has no screen. Where did your `print()` messages go? _____